

REGULATORY HARMONIZATION IN A RESOURCE-LIMITED SETTING: THE WORLD
HEALTH ORGANIZATION COLLABORATIVE PROCEDURE FOR ACCELERATED
REGISTRATION

by

Lori Alquier

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Dedication

To Laurent, who makes everything possible

PREVIEW

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PREVIEW

Abstract

The large numbers of diseases that can only be treated with expensive drugs create economic and societal problems in resource-limited regions. The problem can be exaggerated by burdensome regulatory processes that limit patient access to necessary medicines. Regulatory harmonization of drug approval pathways can potentially reduce the time and costs of registration and thus help to improve patient access to these important medicines. This survey-based research examines the views of National Medicines Regulatory Agencies (NMRAs) and industry professionals on the benefits and barriers associated with a specific regulatory harmonization program, the World Health Organization (WHO) Collaborative Registration Procedure (WHO-CRP) as it is used in Africa. The WHO-CRP is a unique regulatory pathway whose evaluation is particularly useful because the work-sharing and reliance activities of the WHO-CRP program are used across much of Africa. Results reflect the views and experience of 124 participants working in NMRAs and industry. The broadly positive feedback suggests that the primary stakeholders welcome the WHO-CRP program, which not only creates a more efficient pathway for product registration but also achieves its goal of improving the capabilities of less mature NMRAs. The biggest challenges appear to exist outside the specific scope of the regulatory pathway itself. The absence of consistent legislation to facilitate regulatory harmonization, limitations on resources, and continuing challenges of improving regulatory capacity now appear to be the biggest problems to solve. Changes in these elements might have a positive impact on the timeliness and quality of the review and approval processes, not only for the WHO-CRP pathway but for other pathways that depend on harmonization and reliance. The lack of consistent legislation identified in this

study reinforces the conclusions drawn previously about the need for common or consistent legislation. Responses from both NMRA and industry underscore the need for more AU Member States to implement the Model Law if the speed of harmonization is to improve. Once that is achieved, it will facilitate the creation of a self-sustaining infrastructure of interactive NMRAs across the entire continent of Africa that employs work-sharing, which is expected to improve the regulatory capacity of the NMRAs. Improving the visibility and awareness of the program may also be beneficial, particularly if this is coupled with the exploration of a reasonable sliding scale of registration funding. Moving to a self-supporting financial model that supports small, local, and regional production of medicines but would also provide incentives to larger companies outside the region, perhaps by guaranteed timelines and increased transparency, may provide a sustainable future for the program.

Chapter 1. Overview

1.1 Introduction

Healthcare is a basic human right. The Constitution of the World Health Organization (WHO) states that “The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition” (International Health Conference, 2002). About a quarter of the world’s lower- and middle-income countries have explicitly guaranteed specific rights to public health and medical care in their national constitutions (Heymann *et al.*, 2013). Nonetheless, over 2 billion people worldwide do not have sufficient access to essential medicines. Increasing access to high-quality medicines could save 10 million lives each year, including the 4 million people who live in Africa and South-East Asia (Zarocostas, 2018).

Equitable access to medicines is an essential component of good health (Aldis, 2008). Pharmaceutical products are an indispensable component of healthcare systems to prevent, diagnose and treat disease. Still, resource-limited countries are challenged to provide essential medicines of high-quality in a timely manner. Virtually every country provides pharmaceutical regulation and oversight through its own regulatory authority. Those authorities are responsible for regulating the drug development and approval process; however, the effectiveness of this oversight varies. Regulatory authorities differ significantly in their capabilities and, as a result, in their ability to protect their citizens by providing them with safe and effective pharmaceutical products.

Not surprisingly, pharmaceutical regulation has been recognized as a key function to ensure that the drugs to which the public has access will function appropriately, safely

and effectively when needed (Hill and Johnson, 2004). Nonetheless, approximately 75% of National Medicines Regulatory Agencies (NMRAs) globally are limited in their capacity to perform core regulatory functions (Nayyar *et al.*, 2015). According to the WHO regulatory-systems strengthening database, less than 4% of the NMRAs in Africa have a moderate functional capacity (Broojerdi *et al.*, 2020), and the WHO estimates that 90% of NMRAs in Africa have insufficient capacity to undertake basic regulatory functions (Nayyar *et al.*, 2015). Further, the speed at which countries approve pharmaceutical products varies significantly from country to country. A typical delay of 4 to 7 years can be identified from the time in which the first filing submission is made in a high-income country to its approval in Sub-Saharan Africa (Ahonkhai *et al.*, 2016).

Disparate and repetitive regulatory actions in different countries are well-known to cause delays in drug access and have been targeted by regulatory harmonization initiatives around the world (Azatyan, 2020). How best to develop and implement those harmonized approaches is a difficult question to answer. Some have suggested that harmonization initiatives should be coordinated with the work of Stringent Regulatory Authorities (SRAs), a designation used for national drug regulatory authorities, including the United States (US) Food and Drug Administration (FDA), European Medicines Agency (EMA), or Pharmaceuticals and Medical Devices Agency (PMDA) that are members, observers, or associates of the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) (Brennan *et al.*, 2007; Doua and Geertruyden, 2014). Others have suggested that harmonization efforts should be pursued in conjunction with large organizations such as the ICH (Molzon *et al.*, 2011). However, initiatives largely designed using the rules and

experience of highly industrialized economies may require under-resourced countries to implement standards beyond the economic means of their manufacturers or regulators. For that reason, recent harmonization efforts in Africa have favored more locally-driven programs (African Press Organization, 2012; Luthuli and Robles, 2017; Ndomondo-Sigonda and Ambali, 2011).

While all countries in Africa, with the exception of the Sahrawi Arab Democratic Republic, have some type of NMRA (Ndomondo-Sigonda *et al.*, 2017), these agencies vary significantly in their funding, functionality, and capacity (Moran *et al.*, 2011; Wirtz *et al.*, 2017), making a “one size fits all” solution elusive so far. Nevertheless, approximately 85% of countries in Sub-Saharan Africa are involved in one or more initiatives to increase regulatory harmonization (Luthuli and Robles, 2017; Ndomondo-Sigonda *et al.*, 2018). Specifically, partnering countries have been experimenting over the last decade with ways in which they can share the reviews of new pharmaceutical entities and at the same time maintain their rights to decide individually upon the merits of a drug application (Arthur, 2019; Bollyky, Cockburn, and Berndt, 2010). These efforts have resulted in alternatives to standard pathways, often called Facilitated Regulatory Processes (FRPs), to accelerate product development, simplify market authorization application (MAA) submissions, or shorten regulatory reviews. Each program takes a different approach, but all share the goal of reducing the resources needed by both industry and health authorities. Programs such as the Economic Community of West African States (ECOWAS), East African Community (EAC), Southern African Development Community (SADC), and ZAZIBONA initiatives have been set up regionally during that time to capitalize on regional economic organizations and build

upon the framework of regional pharmaceutical regulatory organizations and existing legal tools. Nevertheless, their goals can be more aspirational than realistic.

The World Health Organization Collaborative Procedure for Accelerated Registration (WHO-CRP) is a more recent regulatory harmonization initiative to advocate and plan resource sharing. The program was initially conceived in 2012 to address the increasing backlog of pending applications for drug registration in many low- and middle-income countries. Its goal is to facilitate and accelerate national regulatory approvals by supplying a member NMRA with confidential access to the evidence that a relevant SRA had used to approve and register a pharmaceutical product in its own jurisdiction (with the permission of the Sponsor) (Azatyan, 2019). A pilot program was initially implemented and then followed by a more disseminated final program at the end of 2018. The supra-regional approach used by the WHO to create the newly finalized WHO-CRP program differs from some of its predecessors in that the program is built using flexible tools that allow the underlying infrastructure of national regulatory approval within a country to be used within a structured framework of harmonization.

As regulators on the African continent makes continued efforts to pursue additional harmonization efforts, examination of the strengths and pitfalls of the WHO-CRP program from the perspective of regulators, industry and other key stakeholders can provide valuable learnings to incorporate into future regulatory harmonization efforts.

1.2 Statement of the Problem

Because the limited resources of African regulatory agencies are being expended to support harmonization programs, it is critical to assess whether harmonization initiatives are value-added or resource-draining. Liberti has recently provided an

overview of the FRPs and described significant diversity in the FRPs from the regulatory agencies of resource-limited countries by region (Liberti *et al.*, 2016). Some researchers have analyzed specific harmonization systems, such as the East Africa Community Harmonization Program (EAC) (Dansie, Odoch, and Årdal, 2019), harmonization efforts in West Africa (Kamwanja *et al.*, 2011) and the SADC (Calder, 2016), but these evaluations have not yet extended to the WHO-CRP as an FRP, since the WHO-CRP final program is relatively new compared to some of the other more established regional programs. Caturla Goñi (2016) presented some case studies of experiences that corporations have reported with respect to the WHO-CRP Pilot Program, a precursor to the WHO-CRP final program. These reports suggested that the overall experiences that corporations have had with the pilot program were positive and had the potential to improve the efficiency of participating regulators. To date, no systematic study has evaluated the WHO-CRP in Africa to determine factors specific to these Africa FRPs that might present significant opportunities or barriers to a fully efficient FRP.

1.3 Purpose of the Study

This study evaluated the state of implementation and functionality of the new WHO-CRP program. The goal of this research was to examine the views of stakeholders who are using the finalized WHO-CRP in Africa to determine if the process is useful and represents an overall improvement from the programs that were previously available. It also aimed to identify the perceived barriers pertaining to the use of this pathway and best practices for overcoming these challenges that might inform facilitated regulatory pathways being set up in Africa and other resource-limited settings around the world. Specifically, this research addressed the following research questions:

1. Does the WHO-CRP program meet the needs of the industry professionals and regulators who use the process for product registration in Africa?
2. Does the WHO-CRP program represent an overall improvement over the programs that were previously available for product registration?
3. What are the perceived barriers pertaining to the use of this pathway and positive attributes of the process that might inform facilitated regulatory pathways being set up in Africa and other resource-limited settings around the world?

The research was carried out using the CIPP Framework, a well-established framework developed by Stufflebeam and Coryn (Stufflebeam and Coryn, 2014) that has been in use for over 60 years. The CIPP Framework was used to guide the development of questions that were disseminated to a diverse group of NMRA professionals and key stakeholders using an electronic survey tool. By using an electronic survey tool directed at NMRA professionals and key stakeholders, questions about the best practices and hurdles of the use of the WHO-CRP in Africa were evaluated to assess its perceived effectiveness from the perspective of these key stakeholders.

1.4 Importance of the Study

Significant resources, time, and effort are needed to set up a harmonized regulatory process in a region in which resources for even basic regulatory functions are scarce. By understanding the benefits and challenges of the procedure, regulators can make informed choices about their participation in the program. A better knowledge of the process might help to identify specific areas of concern and dissonance related to the implementation of the WHO-CRP program. Moreover, these results could inform future

harmonization efforts and best practices based on real world experience to ensure that more robust and efficient processes can be put into place in the future.

Companies and consumers may also benefit from this study if it increases regulatory efficiency. By reducing the perceived barriers that exist in planned harmonization programs, the typical delay of 4 to 7 years between the first filing submission in a high-income country and approval in Sub-Saharan Africa may be reduced, resulting in faster deployment of needed medicines to waiting patients in that region. Companies that are contemplating entrance into an African market may be able to develop a better strategy by knowing what barriers might be anticipated and, to some extent, addressed. Further, by reducing the barriers to regulatory approval across Africa, small and mid-sized companies with limited resources may be incentivized to market their products in the region, providing additional treatment options to underserved populations.

1.5 Limitation, Delimitations, Assumptions

This study was delimited to evaluating the WHO-CRP facilitated regulatory pathway that is currently in place on the continent of Africa. It was specific to pharmaceutical products only; diagnostic products, medical devices, and over the counter products were out of the scope of this research. The participants were restricted to individuals who have direct experience with this program within its lifetime in one of two ways, either by using or planning to use the program to register or review products, or by developing or administering the procedures that form the basis of the pathway. The survey was launched in the latter part of 2020, so it provides a snapshot in time that may

not be typical as time passes but can serve as a benchmark against which future programs may be measured.

Several limitations were potential challenges in a survey like this. Surveys are known to be limited by response numbers and rate. In addition to querying other stakeholders, such as health authorities, the intention was to obtain a cross section of pharmaceutical companies based on size and location. This resulted in relatively small numbers in each group, as many companies do not register their products in Africa. Thus, caution is needed when trying to use the small sample as a fair representation of that industry segment. Further, the numbers of individuals with appropriate experience restricted the size of the pool from which participants were drawn, particularly because the WHO-CRP final program has been in place for a relatively short period of time. Many respondents resided in Africa, which made the identification of suitable participants more difficult. In addition, the survey was only provided in English and in French, which created a language limitation, since not all countries in Africa rely on English or French as their primary language. These factors created challenges to secure the cooperation of enough study participants to effectively represent the overall populations.

Because potential participants are often influenced by the length of the survey when choosing to participate, another limitation was the number of questions that I felt could be presented to the survey participants. This placed a limitation on the depth and breadth of the topics that could be covered and the degree to which dropouts occurred. By limiting the number of questions, certain topics may not have been fully explored, and this may constrain the analysis. Each question had to be framed carefully. Using a focus

group to critique the survey helped to ensure better validity, but limitations still exist related to the capabilities and biases of the researcher and the maturity of the program.

The research was based upon my ability to exercise fair balance and objectivity during the conduct of the study and the analysis of the results. To some extent, this bias was identified and controlled by working with faculty who have more experience with survey research and by employing an external focus group of experts to critique the survey for bias and clarity.

While the countries in Africa are geographically and culturally diverse, for this study, I assumed that the views of study participants were determined predominantly by their experience with the process and the political framework under which the process operates, but not cultural or social aspects of their countries. Nevertheless, the cultural and social norms specific to these regions may have influenced the recommendations made as a result of this research. As a result, recommendations from this research may not be entirely applicable to the creation of a facilitated pathway in another region.

Inherent in this research was also the assumption that all respondents were truthful in their evaluations and expressed their own opinions and experience. Therefore, the survey was conducted to ensure anonymity of responses and participants, and this limited the ability to relate specific data to identifiers of the respondent and his/her organization and location.

1.6 Definitions

A list of terms and acronyms used throughout this thesis may be found in Appendix A.

1.7 Organization of the Thesis

The thesis is organized in five chapters. Chapter 1 provides an introduction the study including the problem statement, the purpose, the delimitations, limitations, and importance of the study. It also provides a brief rationale for the approach that will be taken. Chapter 2 provides a more detailed overview of the background and history of the facilitated regulatory pathway programs by presenting some background on the evolution of the facilitated regulatory pathway programs and a literature review representing the current state of knowledge on this topic. Chapter 3 describes the research methodology used for this study. Appendices to this thesis include Appendix A, which includes a list of terms and acronyms used throughout this thesis, and Appendix B, the survey that was distributed.

Chapter 2. Literature Review

2.1 Introduction

Regulatory harmonization is a strategic imperative to increase the access of patients to pharmaceutical products by streamlining regulatory review by NMRAs (Liberti *et al.*, 2018). Harmonization can decrease the amount and costs of research needed to commercialize drug products and can preserve scarce NMRA resources that review and register these drug products. It is not surprising, then, that multiple harmonization initiatives have taken place over the last few decades in different geographical regions and across different epochs of time.

This chapter will trace the evolution of regulatory harmonization over the past 30 years to show the genesis and evolution of harmonization efforts that have influenced the development of such efforts in Africa. The chapter will then describe more specifically the historical development of the WHO-CRP program on which this dissertation will focus.

2.2 Background to the Literature Review

The literature review has three main parts. A brief history of regulatory harmonization related to drug evaluation around the world is followed by a deeper review of existing literature concerning regulatory harmonization amongst African regulatory systems. Finally, literature specifically relating to the WHO-CRP will be examined in detail. As the literature review will show, academic literature becomes sparse as attention is focused on African initiatives.

The database PubMed was used as a starting point with the goal of obtaining scientific publications, books, white papers, and conference presentations on regulatory